

# N-Body Gravitational Simulation

A Python simulation of gravitational N-body dynamics using symplectic integration.

## The physics

Each body feels gravitational force from every other body, integrated with a leapfrog (velocity-Verlet) scheme chosen for its energy-conserving properties over long timescales.

## What's in the repo

`simulate.py` -- core integrator. `plot.py` -- trajectory visualization. A three-body figure-eight orbit as a validation test case.

## Status

Working for small  $N$ ; next step is adaptive timestepping for close encounters.